

CoRoT-1b

R-2463

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_210211 · created 2026-08-23T01:21:05+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459256.7567 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.012
Transit depth (Rp/Rs)^2	2.04 +/- 0.35 [%]
Orbital Inclination (inc)	89.53 +/- 1.78
Airmass coefficient 1 (a1)	1.222 +/- 0.035
Airmass coefficient 2 (a2)	-0.137 +/- 0.011
Scatter in the residuals of the lightcurve fit is	2.85 %
Best Comparison Star	#1 - [381, 29]
Optimal Aperture	2.68
Optimal Annulus	9.09
Transit Duration (day)	0.1139 +/- 0.0018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.143 $\pm$ 0.012 vs 0.1388 (deviation 0.28 σ)
Duration fit vs published	0.1139 d vs 0.1046 d (deviation 4.15 σ)
Mid-time O-C	11.1 min
Depth SNR	9.5
Residual scatter	2.85 %

Light curve

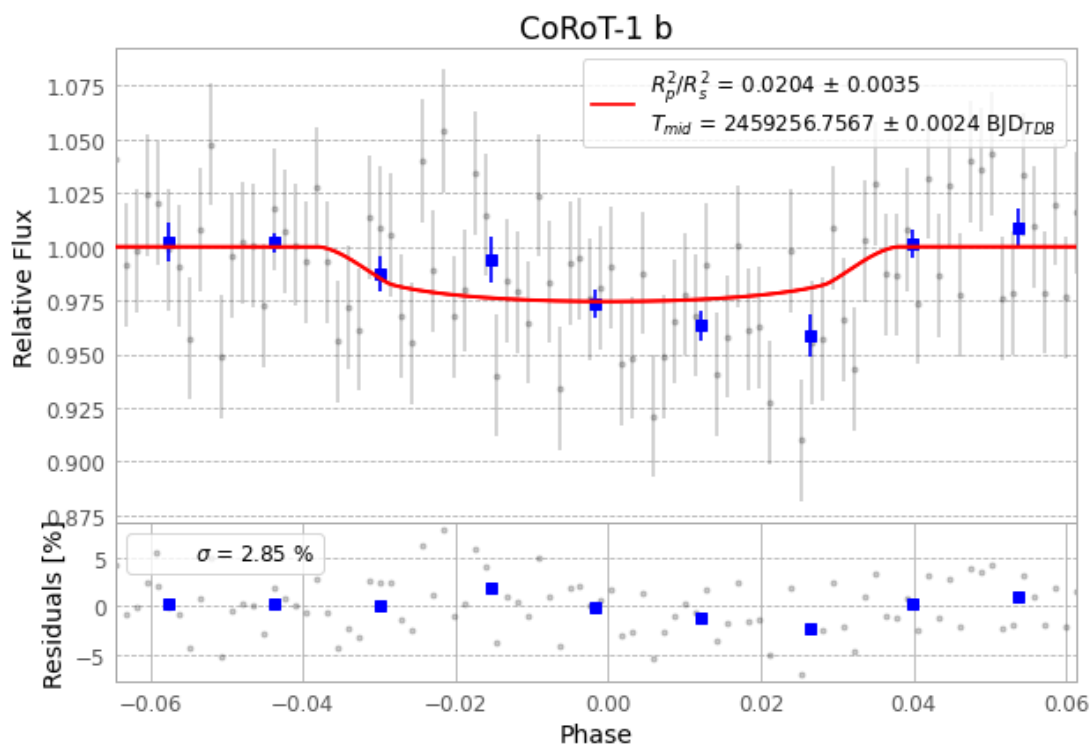


Figure 1 — FinalLightCurve_CoRoT-1 b_2021-02-10.png (SHA-256 8d26e2bc3ab3b7da...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [391, 239], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 195de4bcb6b2eb50...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

92 FITS frames (60 MB), CoRoT-1210211034212.FITS → CoRoT-1210211081512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-210211.log (8cdfb081742d...), FinalLightCurve_CoRoT-1 b_2021-02-10.png (8d26e2bc3ab3...), FinalLightCurve_CoRoT-1 b_2021-02-10.csv (80bda4b54f64...)

Compute resources

Wall time 143.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 01:21Z.